



5.6.5 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the Explorations that I knew were ...	Something I wondered during the Explorations was ...	Something I learned during the Explorations was ...

Unit 5, Lesson 7: Solving Systems of Equations by Graphing



Benchmarks

MA.912.AR.9.1 Given a mathematical or real-world context, write and solve a system of two-variable linear equations algebraically or graphically.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.



Learning Targets

- ☐ I can solve systems of equations by graphing, while noting constraints for a real-world context.
- ☐ I can interpret the solution to a system of equations in a real-world context.
- ☐ I can interpret the coefficients of a given system representing a real-world context.



5.7.1 Warm-Up: Solve the Puzzle

- Determine the value of each figure so that the sum of each row and column match the total value shown.

				172
				136
				136
108	144	108	84	

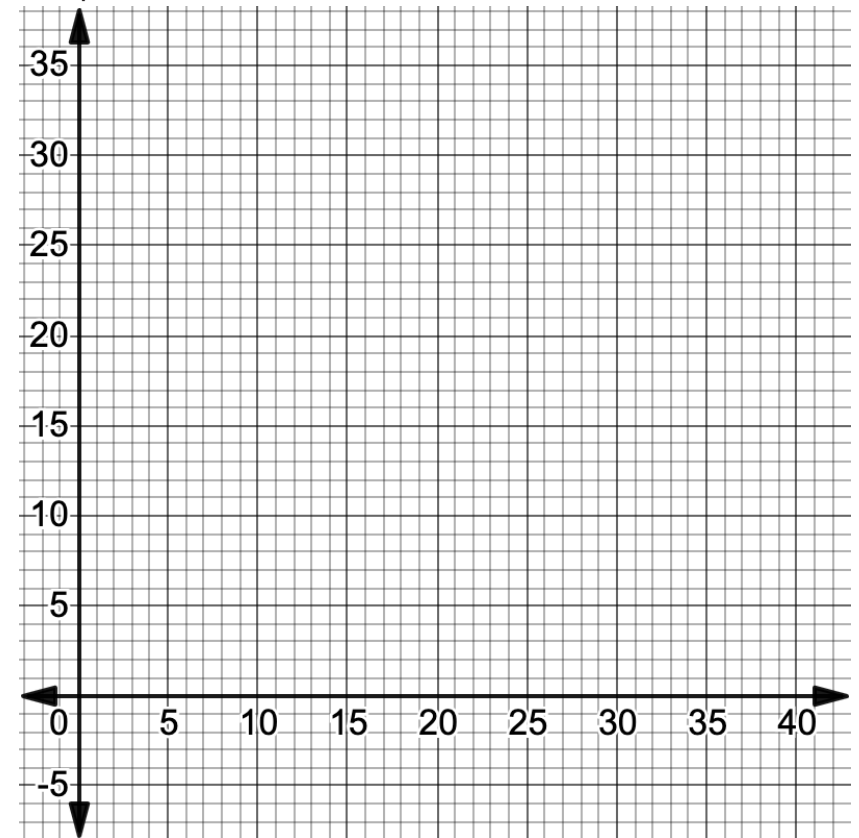
- To solve the puzzle, explain which figure you started with and how you were able to find its value.
- How did you find the values of the other figures after determining the value of the first figure?



5.7.2 Collaboration: Solving Systems of Equations by Graphing

- Use the system of equations $\begin{cases} y = 0.6x + 5 \\ y = 0.3x + 14 \end{cases}$ to complete the following.

- Graph the system of equations on the coordinate plane.



- Complete the statement.

The point (____, ____) is the solution to the system.

2. The system of equations from the previous question can be used to represent the following situation.

Many businesses use Google to track trends to help them make business decisions. A value of 100 represents peak popularity of a search term. A marketing analyst found the following search trends when comparing interest in pho and ramen through search trends in the U.S. since 2010.

- In January 2010, pho had a search interest value of 14, which increased by an average of 0.3 each month until March of 2021.
- In January 2010, ramen had a search interest value of 5, which increased by an average of 0.6 each month until March of 2021.

- a. Match each search term to the equation that could be used to model its trend from January 2010 to March 2021.

Search Term	Equation
pho	○ $y = 0.6x + 5$ ○ $y = 0.3x + 14$
ramen	○ $y = 0.6x + 5$ ○ $y = 0.3x + 14$

- b. Discuss with your partner what the solution to the system found in the previous question means in this context. Then, summarize your discussion.

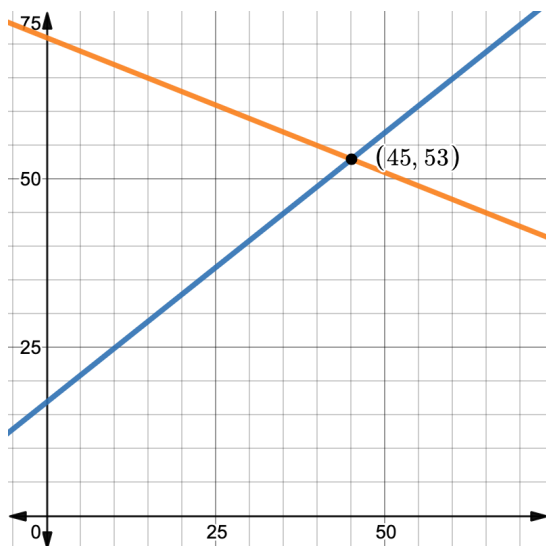
- c. Work with your partner to explain how negative x values relate to the situation.

- d. Discuss with your partner any possible constraints on the x -values in this context.



5.7.3 Guided Instruction: Solving Systems of Equations by Graphing and Interpreting Solutions

1. The graph represents the trends in the percentage of U.S. adults who read e-books versus print books each year since 2011.



- a. Let s represent the number of years since 2011 and p represent the total number of people using the reading format. Match the situation with the appropriate equation and line from the system shown on the graph.

Situation	Equation	Graph
In 2011, 17 percent of the U.S. adults surveyed read e-books, and the percentage increased $\frac{4}{5}\%$ each year after.	☐ $p = -0.4s + 71$ ☐ $p = 0.8s + 17$	☐ Orange line ☐ Blue line

In 2011, 71 percent of the U.S. adults surveyed read print books, and the percentage decreased $\frac{2}{5}\%$ each year after.

- ☐ $p = -0.4s + 71$
☐ $p = 0.8s + 17$

- ☐ Orange line
☐ Blue line

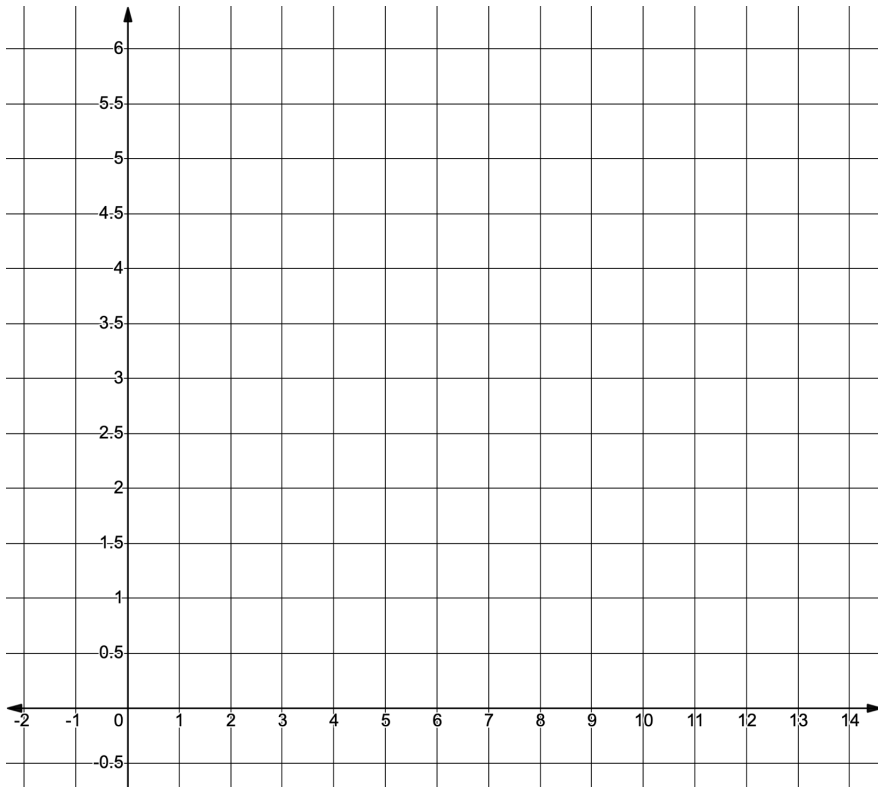
- b. If the trends continue, what year will e-books and print books have the same number of readers?
- c. What percentage of the U.S. adult population is predicted to be using each reading format at that time?
- d. The variable s represents the years since 2011. If $s = -2$, what year is being represented?
- e. Discuss with your partner any real-world constraints that may impact these trends.



5.7.4 Guided Practice: Solving Systems by Graphing and Interpreting Solutions

1. Levi is comparing the average daily precipitation in millimeters (mm) of Calhoun County, Florida to the entire state of Florida. Based on the data he collected from 1999 to 2010, Levi uses the equation $a = \frac{1}{5}w + 3.9$ to represent the precipitation in Calhoun County, FL, and $a = \frac{1}{5}w + 3.4$ to represent the precipitation in the state of Florida. In both equations in the system $\begin{cases} a = \frac{1}{5}w + 3.9 \\ a = \frac{1}{5}w + 3.4 \end{cases}$, a represents the average daily rainfall, in mm, and w represents the number of years since 1999.

a. Graph the system of equations.



- b. What does the slope of the lines represent in the given context?
- c. What was the total average rainfall for Calhoun County in 2003?
- d. Explain whether or not Calhoun County typically has more or less precipitation than the state.

e. Complete the statements.

The two equations in the system are ☐ intersecting ☐ parallel

lines with ☐ the same slope, ☐ different slopes, so they ☐ will ☐ will not

intersect. Therefore, there ☐ is ☐ is not a year when

Calhoun County had the same daily average

precipitation as the state of Florida. There

☐ is one solution ☐ is no solution ☐ are infinitely many solutions

to this system.

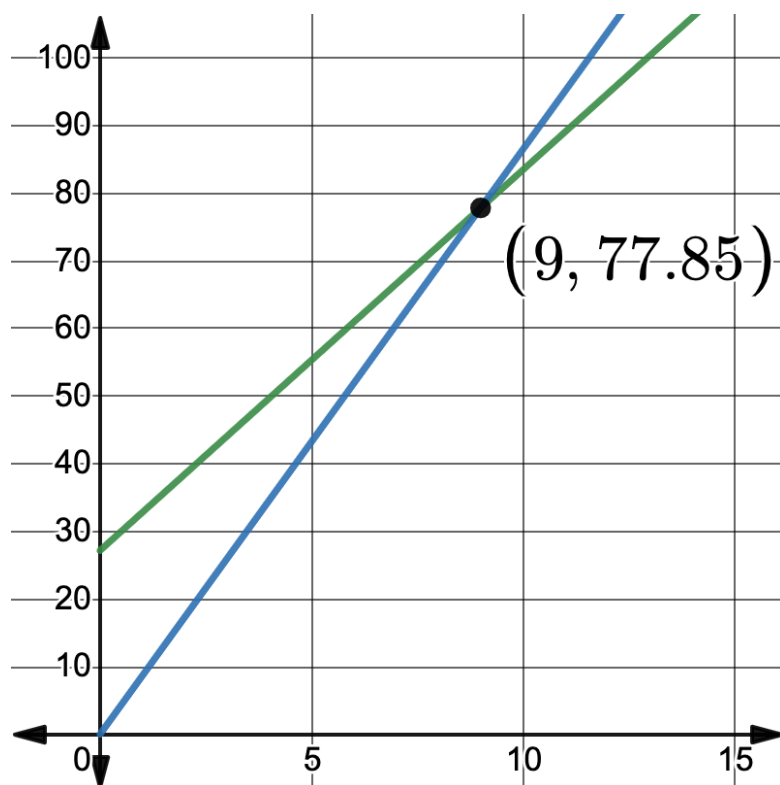




5.7.5 Try It: Solving Systems by Graphing and Interpreting Solutions

1. Remy just started working at the local salon. She is paid \$5.63 per hour and earns, on average, \$27.18 dollars in tips each day. Suki just started working at Publix. Since she does not receive tips, minimum wage for her is \$8.65 an hour. The system of equations $\begin{cases} m = 5.63h + 27.18 \\ m = 8.65h \end{cases}$ can be used to represent the amount of money earned, m , each day for working h hours.

The graph of the system is shown.



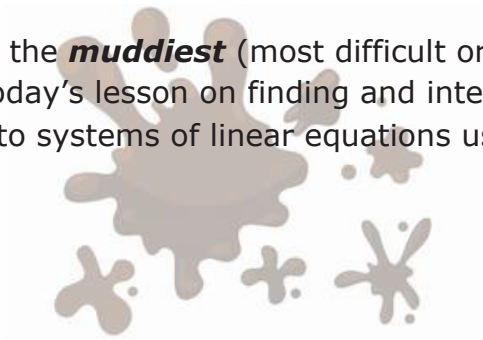
- a. Discuss with your partner why the graphs do not include negative h and m values.

- b. Label each line with the name of the person whose earnings it represents.
- c. What is the solution to this system of equations?
- d. Interpret the meaning of the solution in the context.
- e. Discuss with your partner the circumstances under which either Remy or Suki would earn more money per day. Then, summarize your discussion.



5.7.6 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on finding and interpreting solutions to systems of linear equations using graphs?



Unit 5, Lesson 8: Solving Systems of Equations by Substitution



Benchmarks

MA.912.AR.9.1 Given a mathematical or real-world context, write and solve a system of two-variable linear equations algebraically or graphically.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.



Learning Targets

- ☐ I can solve systems of equations by substitution.
- ☐ I can write a system of equations from a real-world context and solve by substitution.
- ☐ I can interpret the solution to a system of equations in a real-world context.